TRACK: UX/UI DESIGN · WEEK 6 · BUILD+

UX-06B: Component Variant Matrix Generator

Combinatorial Attribute Axes · 156 Permutations · Auto-Layout Specs · Production Code Mapping

Candidate: K. M. Siam
System: FlyCore v1.0

CORE COMPONENTS	TOTAL PERMUTATIONS	FIGMA PLUGIN LOGIC	CODE TRANSLATION
Button, Input, Card	156 Generated Variants	DeepSeek Combinatorial Engine	Figma ↔ React ↔ Tailwind

1. Attribute Axes & Variant Counts

72
Button Variants (4 × 3 × 6)

48
Input Field Variants (4 × 2 × 6)

36
Card Variants (3 × 2 × 2 × 3)

2. Variant Naming Convention & Code Translation Rules

```
// Direct 1:1 Deterministic Mapping Contract:
Figma Slash: Button / Primary / Large / Hover
React Prop: <Button hierarchy="primary" size="lg" state="hover" />
Tailwind: className="btn btn-primary btn-lg hover:bg-indigo-700"

Figma Slash: Input / Text / Medium / Error
React Prop: <Input type="text" size="md" state="error" helperText="Invalid SKU format" />
Tailwind: className="input input-md border-red-500 bg-red-50 focus:ring-red-500"

Figma Slash: Card / Metric / Desktop / Dark / Hover
React Prop: <Card domain="metric" breakpoint="desk" theme="dark" state="hover" />
Tailwind: className="card card-metric dark:bg-slate-900 hover:shadow-md"
```

3. Button Variant Matrix (Sample of 72 Combinations)

Hierarchy	Size	State	Design Token (Fill / Text)	Figma Component Slash Name
Primary	Large (48px)	Default	`action-primary-default` (#4F46E5) / White	`Button / Primary / Large / Default`
Primary	Large (48px)	Hover	`action-primary-hover` (#4338CA) / White	`Button / Primary / Large / Hover`
Primary	Large (48px)	Focused	`action-primary-default` + 2px focus ring	`Button / Primary / Large / Focused`
Primary	Large (48px)	Disabled	`action-primary-disabled` (#C7D2FE) / Muted	`Button / Primary / Large / Disabled`
Secondary	Large (48px)	Default	`surface-default` (#FFFFFF) / Slate-200 border	`Button / Secondary / Large / Default`
Destructive	Large (48px)	Default	`feedback-danger-surface` (#DC2626) / White	`Button / Destructive / Large / Default`
Ghost	Large (48px)	Default	Transparent (0% fill) / Slate-900 text	`Button / Ghost / Large / Default`

4. Input Field Variant Matrix (Sample of 48 Combinations)

Input Type	Size	State	Border & Background	Figma Component Slash Name
Text Input	Medium (40px)	Default	`border-default` (#E2E8F0) / White	`Input / Text / Medium / Default`
Text Input	Medium (40px)	Focused	`border-focus` (#6366F1 2px ring)	`Input / Text / Medium / Focused`
Text Input	Medium (40px)	Error	`feedback-danger-text` (#B91C1C) / Red-50	`Input / Text / Medium / Error`
Counter	Medium (40px)	Default	`border-default` / Tabular figures	`Input / Counter / Medium / Default`
Dropdown	Medium (40px)	Default	`border-default` / Chevron-down icon	`Input / Dropdown / Medium / Default`
Search Bar	Medium (40px)	Default	`border-default` / Search icon leading	`Input / Search / Medium / Default`

5. Card Variant Matrix (36 Multi-Theme Combinations)

Domain	Breakpoint	Theme	State	Figma Component Slash Name
Metric Card	Desktop (1440px)	Light Mode	Static	`Card / Metric / Desktop / Light / Static`
Metric Card	Desktop (1440px)	Light Mode	Hover	`Card / Metric / Desktop / Light / Hover`
Metric Card	Desktop (1440px)	Dark Mode	Static	`Card / Metric / Desktop / Dark / Static`
Batch-Order	Tablet (1024px)	Light Mode	Static (54px touch)	`Card / Batch-Order / Tablet / Light / Static`
Batch-Order	Tablet (1024px)	Dark Mode	Static (54px touch)	`Card / Batch-Order / Tablet / Dark / Static`
SKU-Detail	Desktop (1440px)	Light Mode	Static	`Card / SKU-Detail / Desktop / Light / Static`

Master Design System File (View Access): https://www.figma.com/file/r7K8x9pL4mN2qS/FlyRank-UX-UI-Design-Capstone?type=design&node-id=0%3A1&mode=design&t=review-access
GitHub Repository: https://github.com/KMSiam/Flyrank-Internship-UX-UI-Design